An introduction to reinforcement learning from human feedback and post-training

SALA 2026

Nathan Lambert

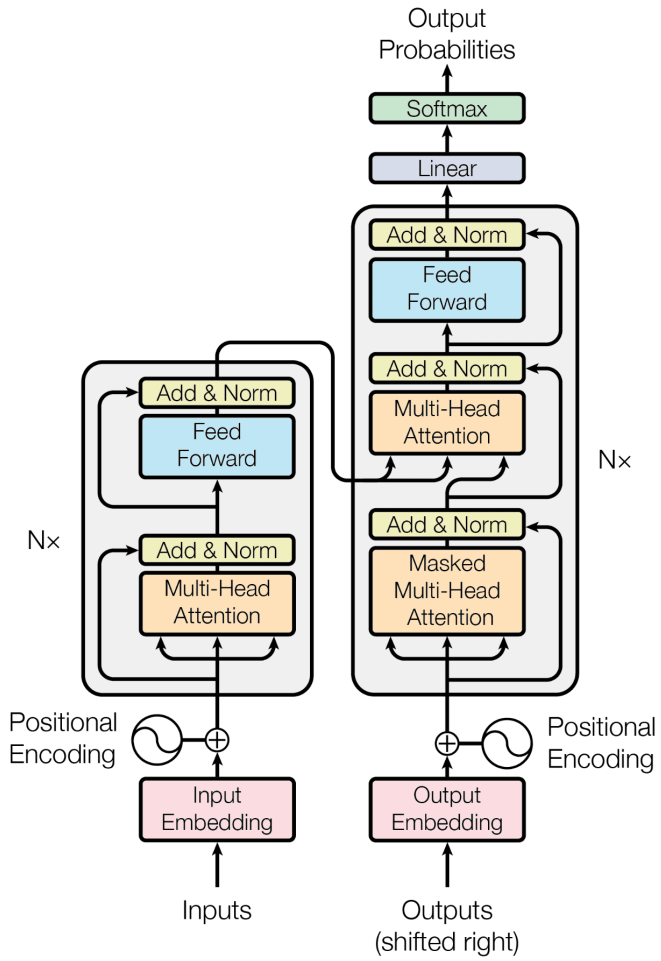
Quito, Ecuador
11 March 2026

A cursory overview of RLHF, RLVR, and modern post-training recipes for language models.

What is a language model?

Core properties:

- A language model assigns probabilities to text.
- Chunks of words are broken down as **tokens**, which are the internal representation of the model.
- Given previous tokens, it predicts the next token. Repeating this produces a completion one step at a time (this is called **autoregressive**).



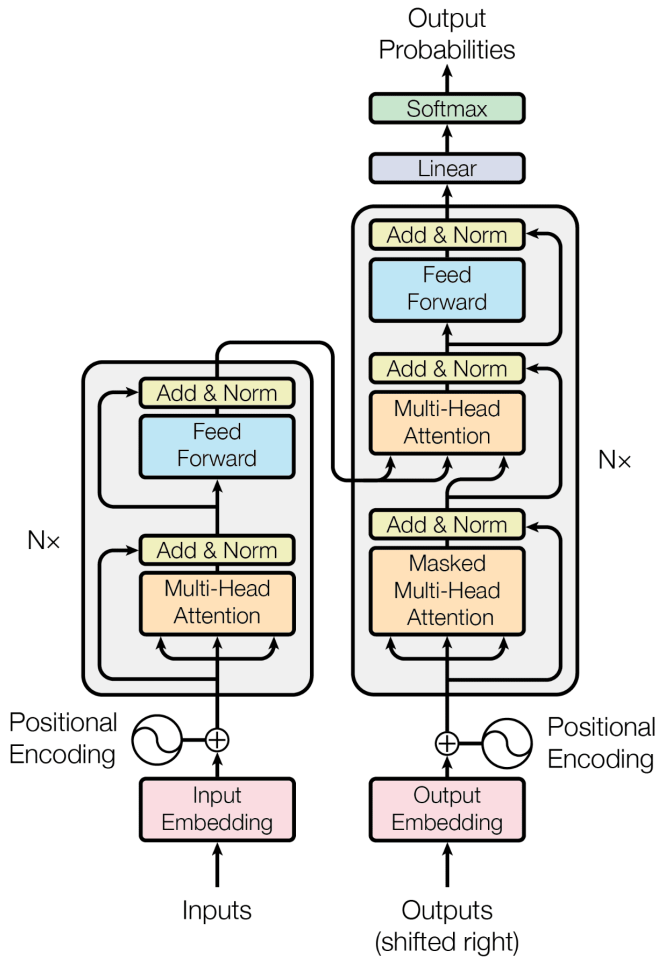
The original model architecture diagram for the Transformer. 2017.

Vaswani et al., 2017

What is a (modern) language model?

Modern language models:

- Have billions to trillions of parameters.
- Largely downstream of the Transformer architecture, which popularized the use of the **self-attention** mechanism along with fully-dense layers.
- Predict and work over much more than text: Gemini and ChatGPT work with images, audio, and video.

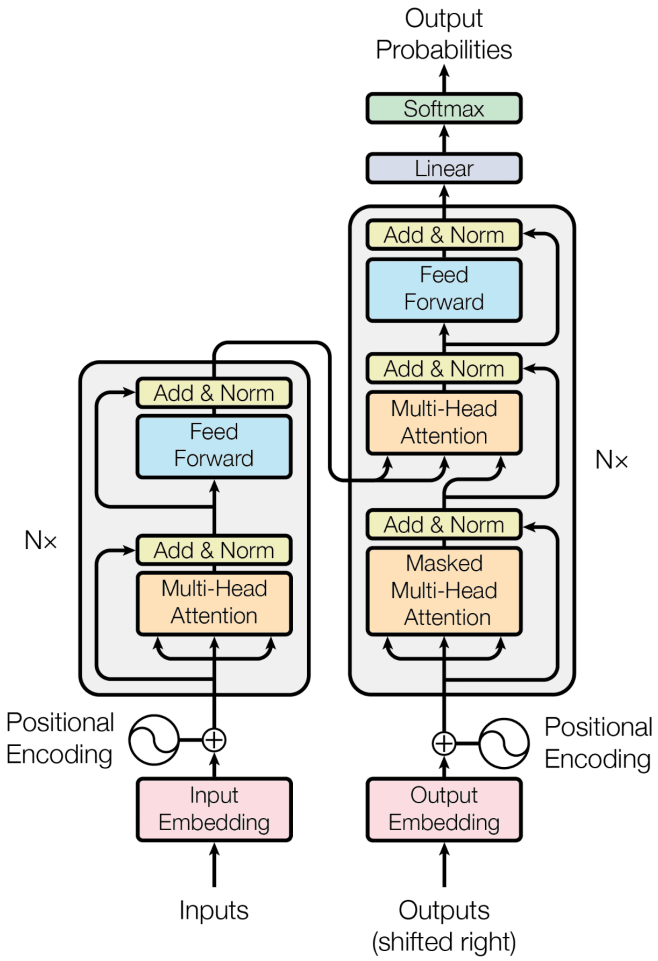


The original model architecture diagram for the Transformer. 2017.

Vaswani et al., 2017

2017: The Transformer is born

- **2017:** the Transformer is born

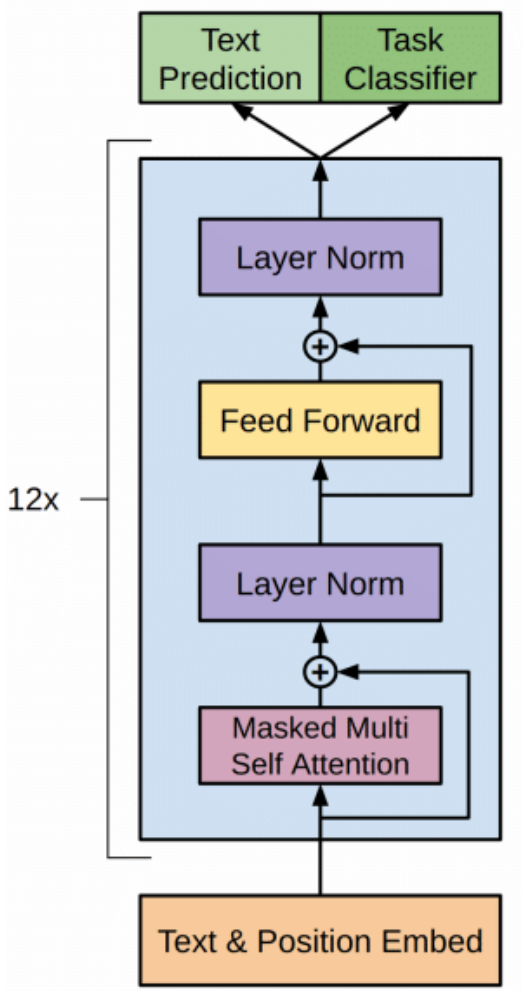


The original model architecture diagram for the Transformer. 2017.

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2018: GPT-1, ELMo, and BERT

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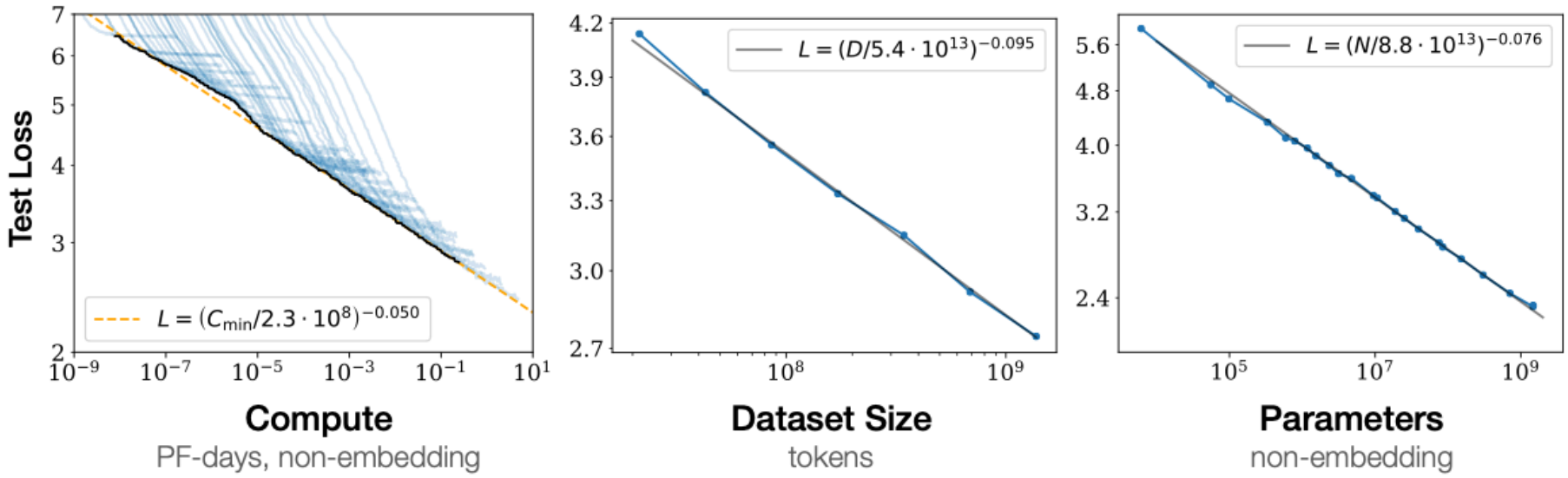


The language model architecture for GPT-1. 2018.

Devlin et al., 2018; Peters et al., 2018; Radford et al., 2018

2019: GPT-2 and scaling laws

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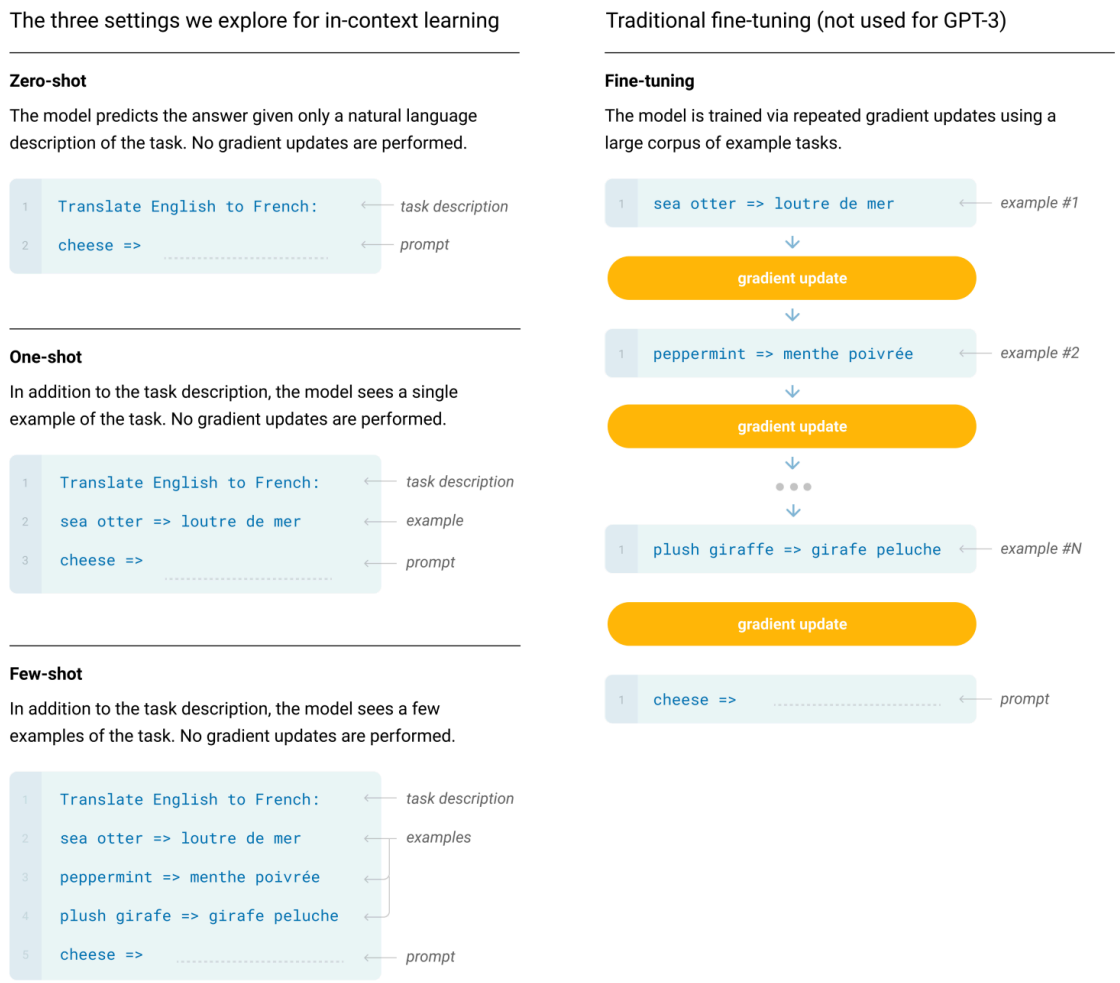


The famous scaling laws plots. 2020.

Kaplan et al., 2020; Radford et al., 2019

2020: GPT-3's surprising capabilities

- 2017: the Transformer is born
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- **2020:** GPT-3's surprising capabilities



GPT-3 was known for expanding the idea of in-context learning and few-shot prompting. Screenshot from the paper.

Brown et al., 2020

2021: Stochastic Parrots

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- 2020: GPT-3's surprising capabilities
- **2021:** Stochastic Parrots



On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?

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ABSTRACT

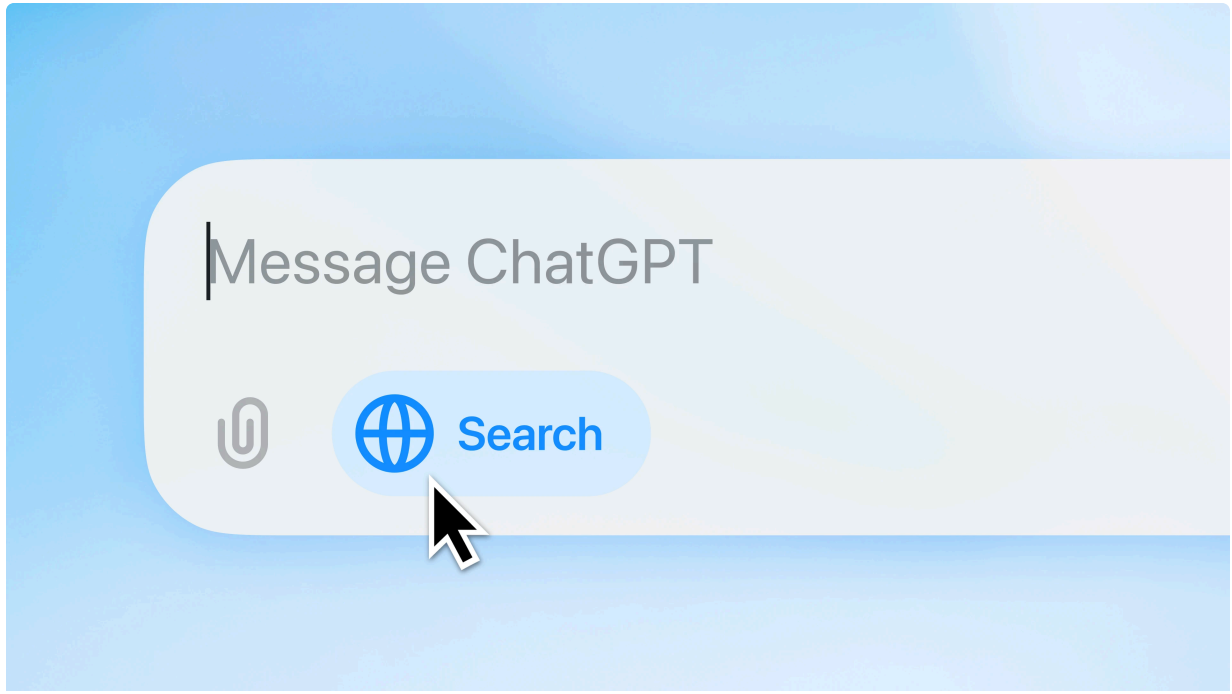
The past 3 years of work in NLP have been characterized by the development and deployment of ever larger language models, especially for English. BERT, its variants, GPT-2/3, and others, most recently Switch-C, have pushed the boundaries of the possible both through architectural innovations and through sheer size. Using these pretrained models and the methodology of fine-tuning them for specific tasks, researchers have extended the state of the art on a wide array of tasks as measured by leaderboards on specific benchmarks for English. In this paper, we take a step back and ask: How big is too big? What are the possible risks associated with this technology and what paths are available for mitigating those risks? We provide recommendations including weighing the environmental and financial costs first, investing resources into curating and carefully documenting datasets rather than ingesting everything on alone, we have seen the emergence of BERT and its variants [39, 70, 74, 113, 146], GPT-2 [106], T-NLG [112], GPT-3 [25], and most recently Switch-C [43], with institutions seemingly competing to produce ever larger LMs. While investigating properties of LMs and how they change with size holds scientific interest, and large LMs have shown improvements on various tasks (§2), we ask whether enough thought has been put into the potential risks associated with developing them and strategies to mitigate these risks.

We first consider environmental risks. Echoing a line of recent work outlining the environmental and financial costs of deep learning systems [129], we encourage the research community to prioritize these impacts. One way this can be done is by reporting costs and evaluating works based on the amount of resources they consume [57]. As we outline in §3, increasing the environmental and financial costs of these models doubly punishes marginalized com-

Bender et al., 2021

2022: ChatGPT

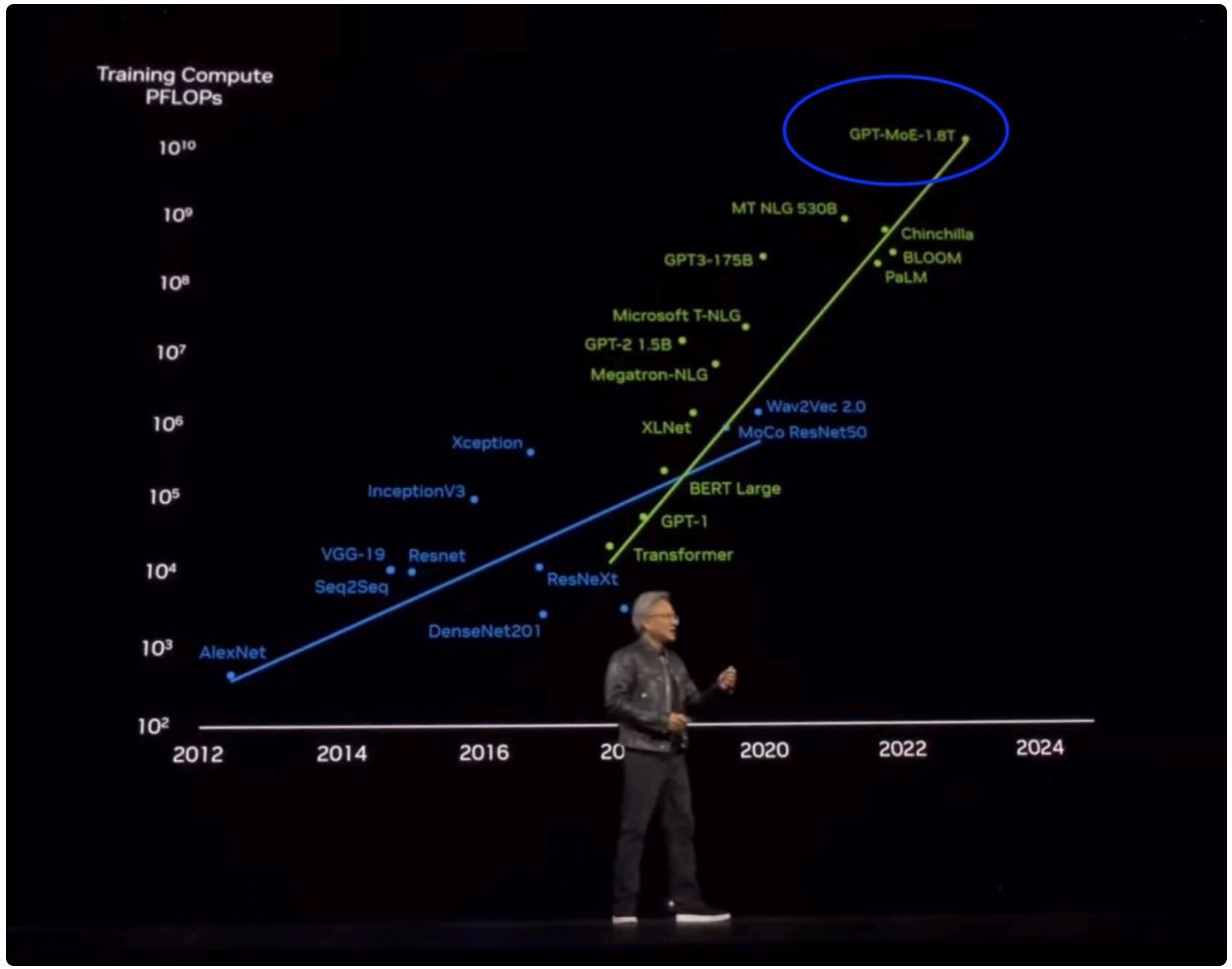
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- 2021: Stochastic Parrots
- **2022:** ChatGPT



OpenAI, 2022

2023: GPT-4 and frontier-scale

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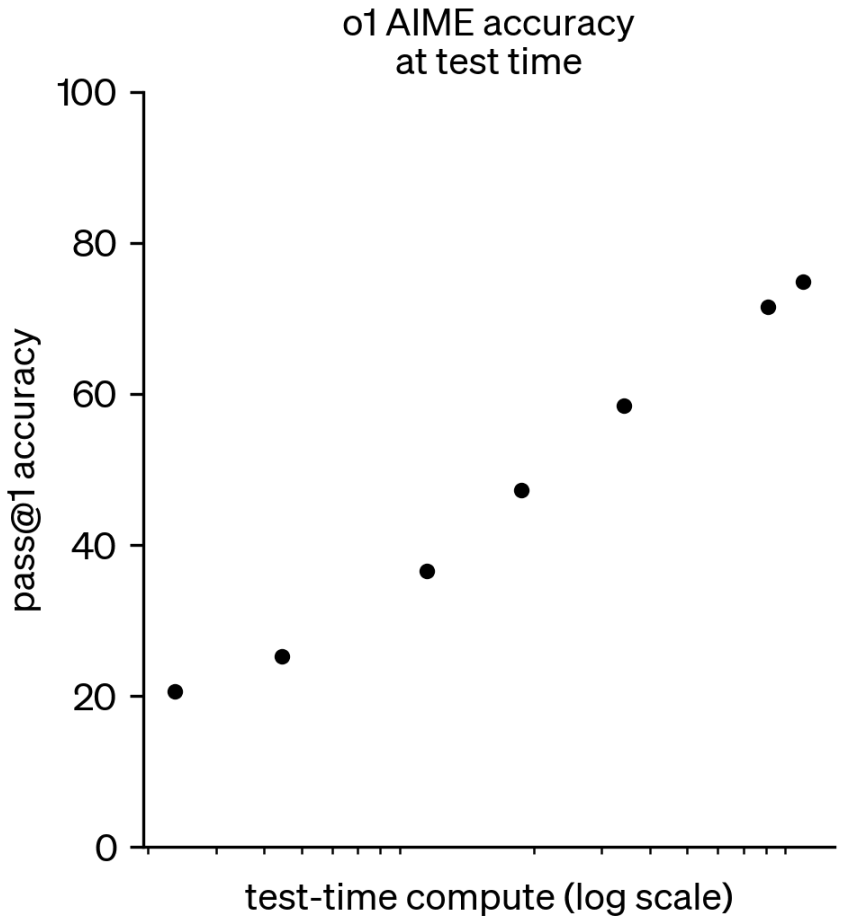


An image where NVIDIA CEO Jensen Huang supposedly leaked that GPT-4 was a roughly 2T-parameter MoE model.

OpenAI, 2023

2024: o1 and reasoning models

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- **2024:** o1 and reasoning models



The famous test-time scaling plot from OpenAI's o1 announcement.

OpenAI, 2024

2025: o3, Claude Code, and agents

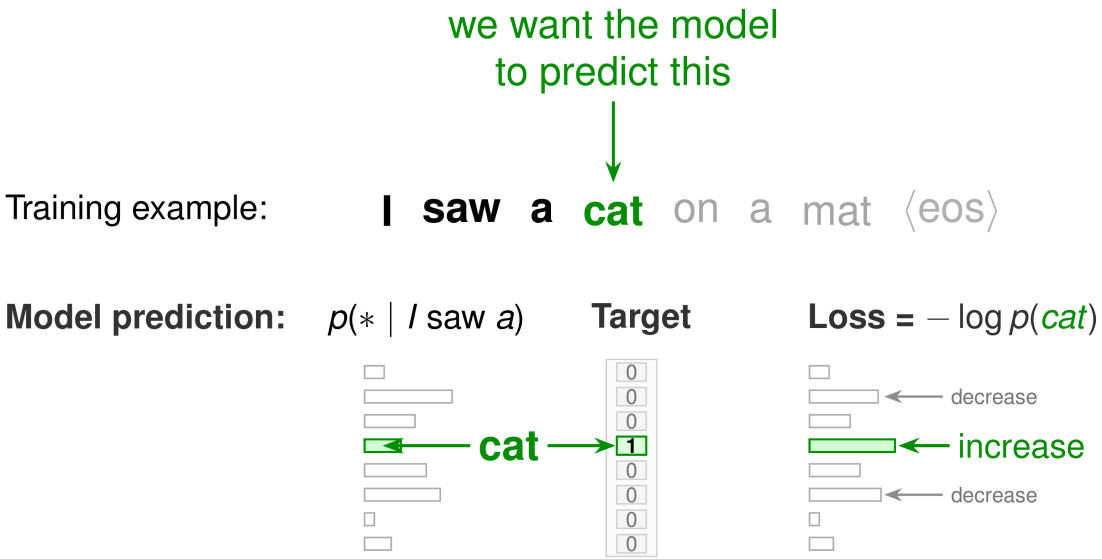
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Anthropic, 2025; OpenAI, 2025; Team, 2025

Pretraining: next-token prediction

- Train on trillions of tokens of text from the web, books, code, and documents
  - Models are often trained on 5-50+ trillion tokens
  - 1T of text tokens is about 3-5 TB of data
  - Labs gather and filter 10-20X more data than is used for the model
  - Total data funnel targeted for models is on the order of petabytes
- Objective: predict the next token in each sequence
- Result: Incredible, flexible, useful models



A base model completes text

After pretraining we are left with a glorified autocomplete model, for example:[1]

USER

The president of the United States in 2006 was

ASSISTANT (Llama 3.1 405B Base)

George W. Bush, the governor of Florida in 2006 was Jeb Bush, and John McCain was an Arizona senator in 2006...

1: Base models are also becoming more flexible through midtraining and better data mixtures.

Post-training makes it answer like a chatbot

The earliest forms of modern post-trained (or RLHF-tuned) models shifted the continuation format to always conforming to the “answering a question style.” An example of what early conversational models looked like is below:

USER

The president of the United States in 2006 was

ASSISTANT (Tülu 3 405B)

George W. Bush was the president of the United States in 2006. He served two terms in office, from January 20, 2001, to January 20, 2009.

ChatGPT was when RLHF made the models even easier to use

Model responses evolved quickly to have:

- Better **format**: direct, conversational answers
- Better **style**: helpful, concise, Markdown, etc.
- Better **product**: people could actually use it every day

USER

I'm giving a talk on RLHF tomorrow. Can you help me structure it?

ASSISTANT

Absolutely — here's a simple structure:

1. Start with the basics

- What a language model is
- Why next-token prediction is not enough

2. Explain post-training

- Instruction tuning
- Preference tuning / RLHF
- RLVR for verifiable tasks

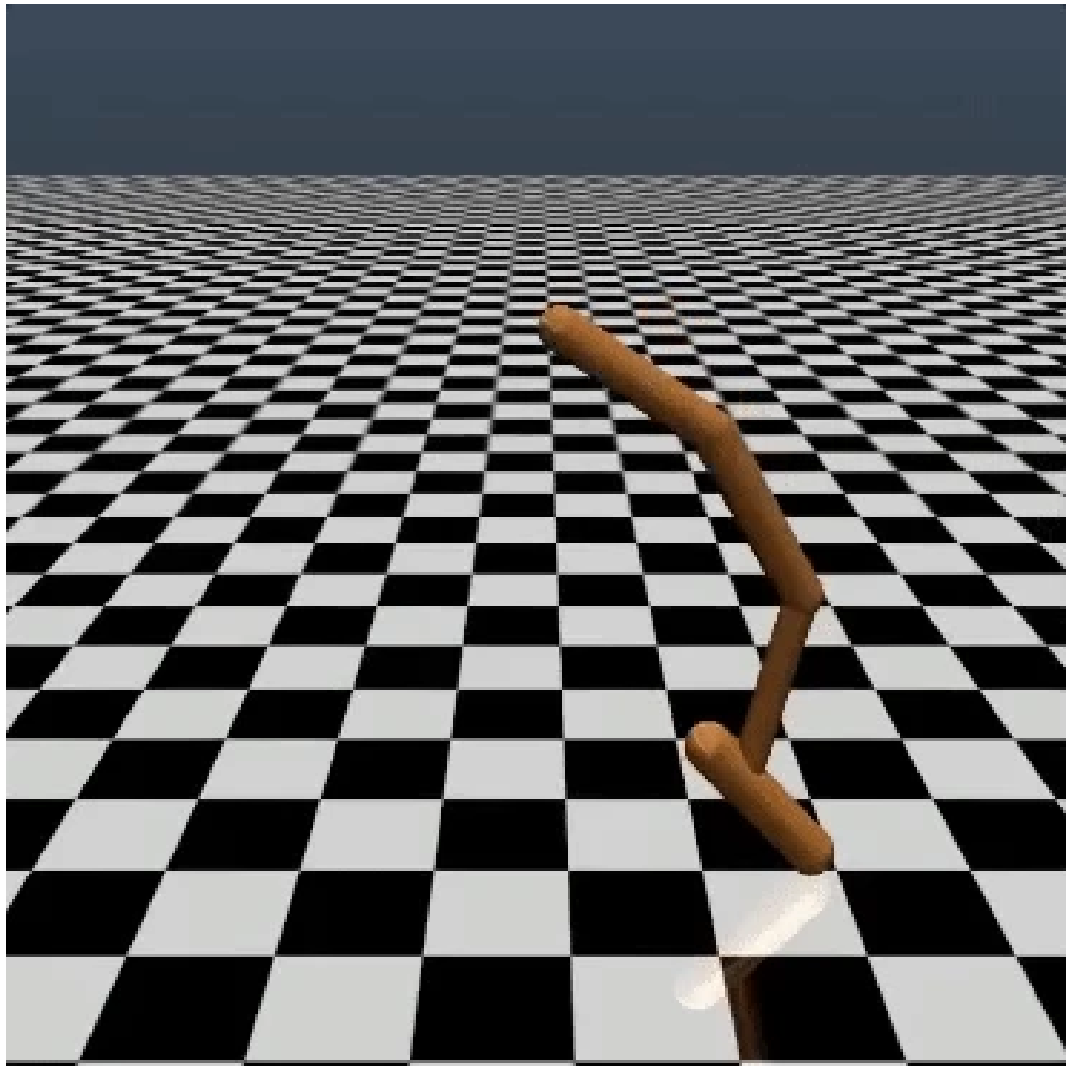
3. End with why it matters

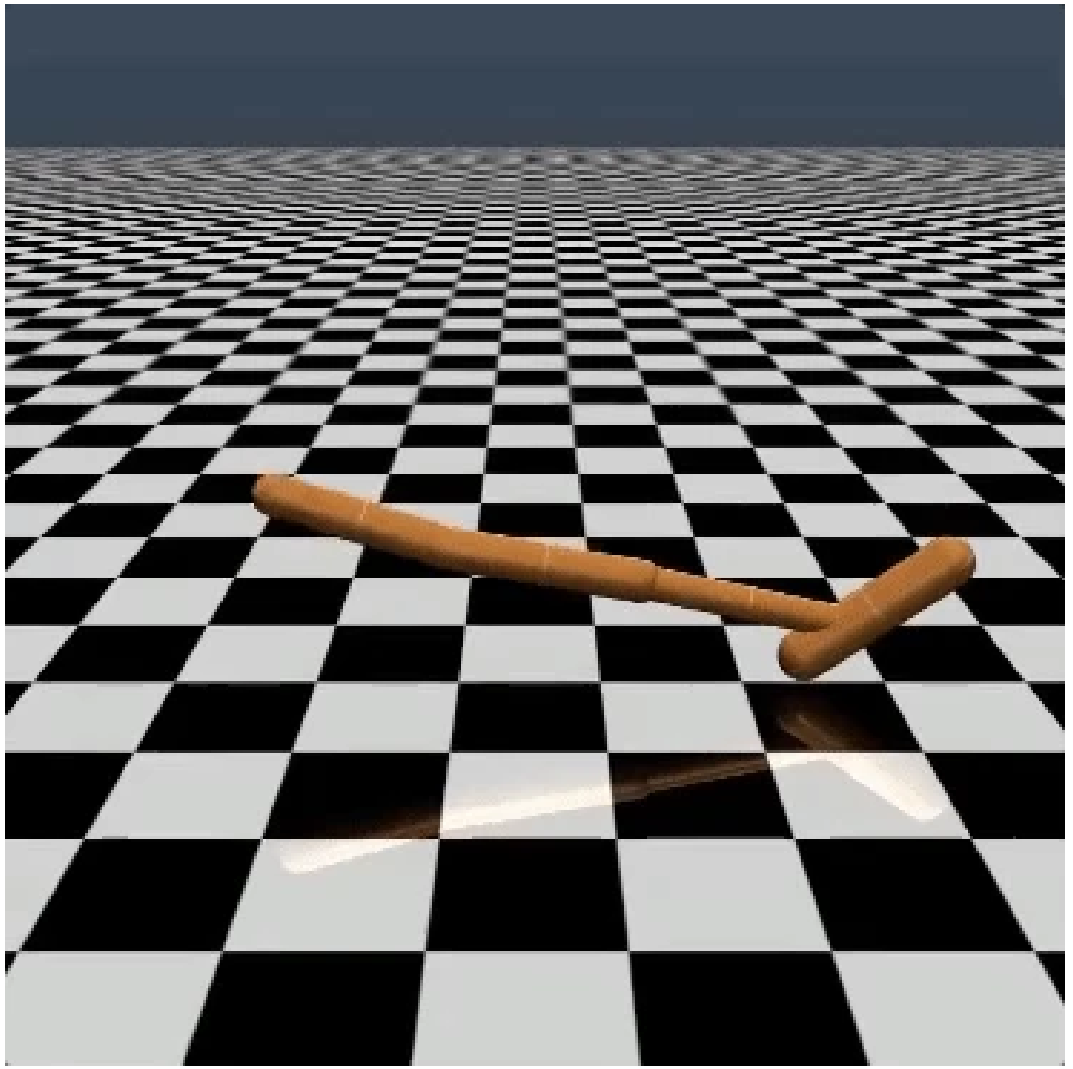
- Better products like ChatGPT
- Better reasoning in models like o1

...

So what is reinforcement learning from human feedback (RLHF) anyways?

Which is the better backflip?





Christiano et al., 2017

Why did people make RLHF?

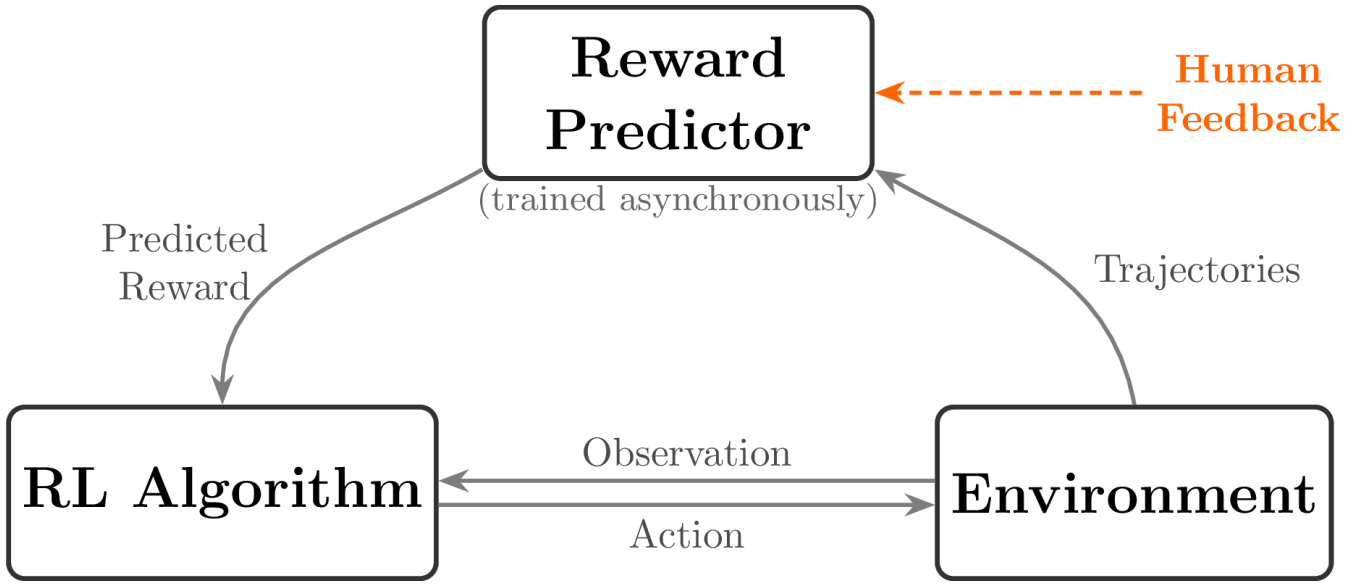
- Many objectives are easy for humans to **judge**, but hard to write as an exact reward function
- In language models, what we want is often implicit: **follow intent**, be **helpful**, be **harmless**
- Pretraining optimizes **next-token prediction**, not assistant behavior
- Preference comparisons turn those human judgments into a scalable training signal

RLHF lets us optimize for behavior we can **evaluate**, even when we cannot easily **specify** the reward.

Bai et al., 2022; Christiano et al., 2017; Ouyang et al., 2022; Ziegler et al., 2019

RLHF before language models

- **TAMER** (Knox & Stone, 2008) — humans score agent actions to learn a reward
- **Christiano et al. 2017** — RLHF on Atari trajectory preferences
- **Ziegler et al. 2019** — first RLHF on language models



A recreation of the system diagram from Christiano et al. 2017.

Christiano et al., 2017

RLHF for language models: compare two completions

USER
Explain the moon landing to a 6-year-old.

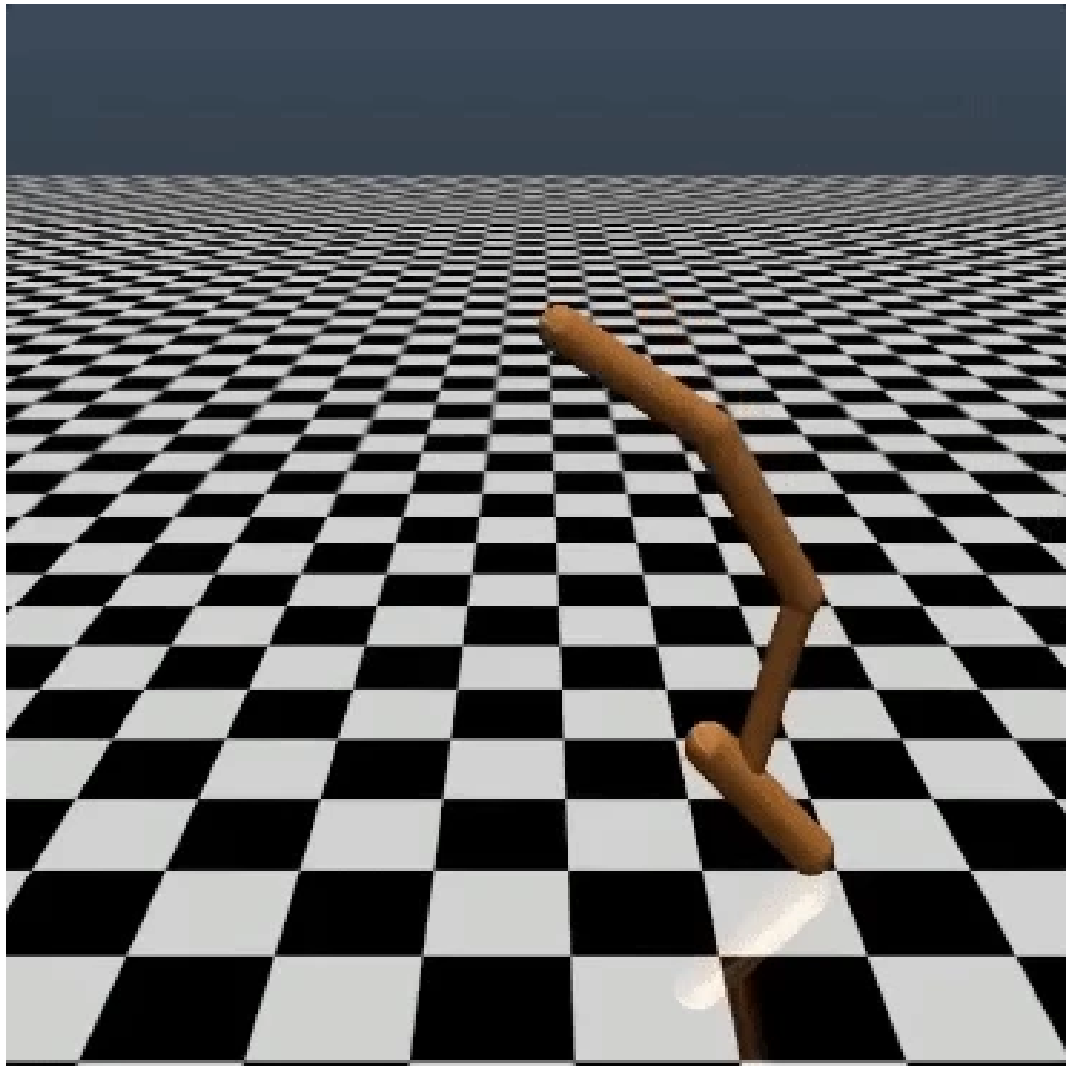
ASSISTANT (Completion A)
The Apollo program culminated in a successful lunar landing in 1969. Astronauts used a spacecraft to descend to the moon's surface and collect samples before returning to Earth.

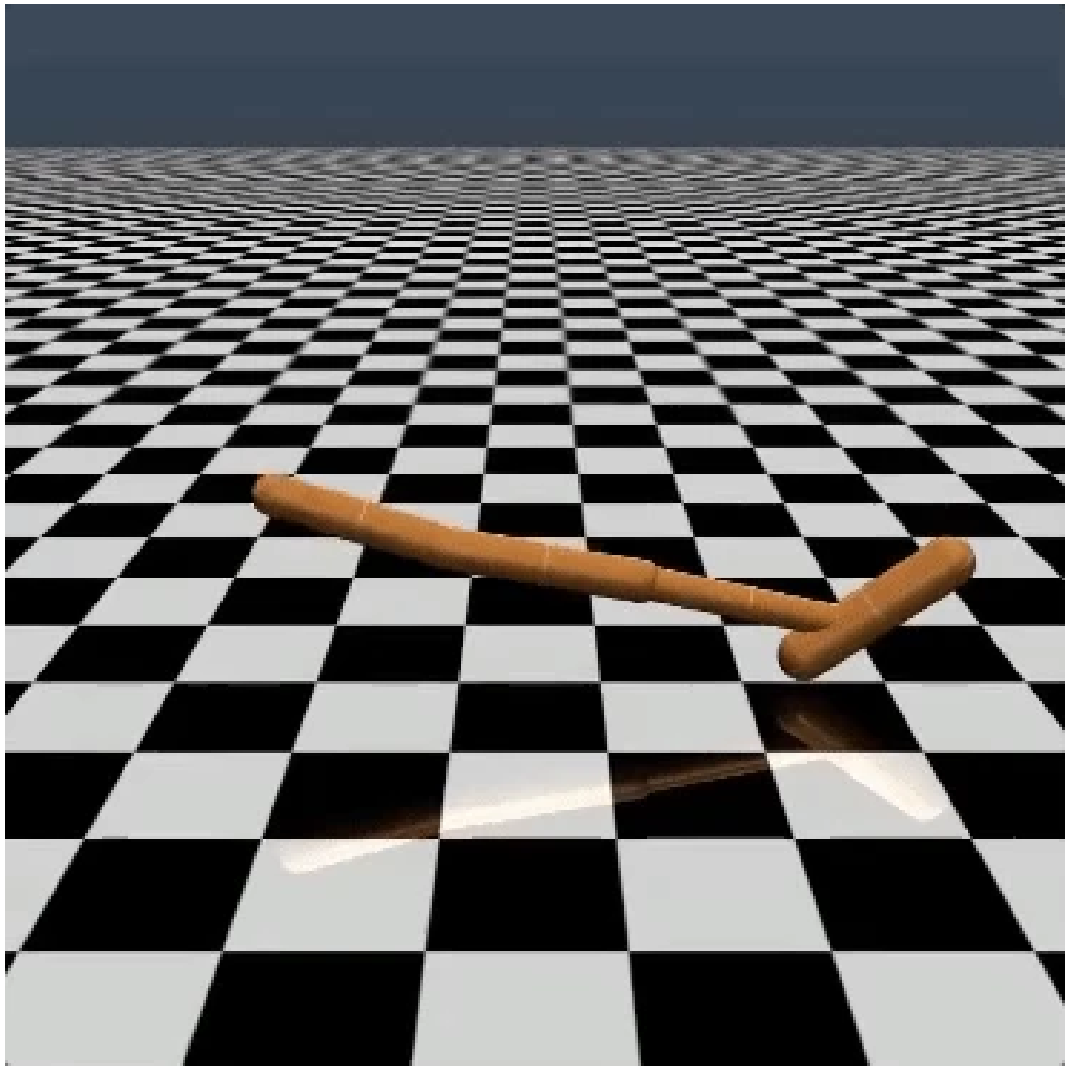
USER
Explain the moon landing to a 6-year-old.

ASSISTANT (Completion B)
People built a special rocket to go to the moon. Two astronauts landed there, walked around, and came home safely to tell everyone what they saw.

Ziegler et al., 2019

Left: Human feedback; Right: Hand-designed reward function





Christiano et al., 2017

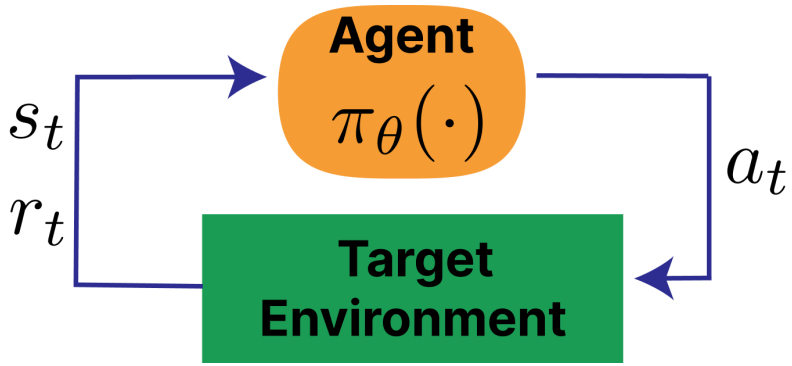
Classical RL

A reinforcement learning problem is often written as a **Markov Decision Process (MDP)**:

- state space $\mathcal{S}$, action space $\mathcal{A}$
- transition dynamics $P(s_{t+1} \mid s_t, a_t)$
- reward function $r(s_t, a_t)$ and discount γ
- optimize cumulative return over a trajectory

$$\text{MDP}\ (\mathcal{S}, \mathcal{A}, P, r, \gamma)$$

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{T} \gamma^t r(s_t, a_t) \right]$$



Sutton & Barto, 2018

Classical RL vs. RLHF

Classical RL

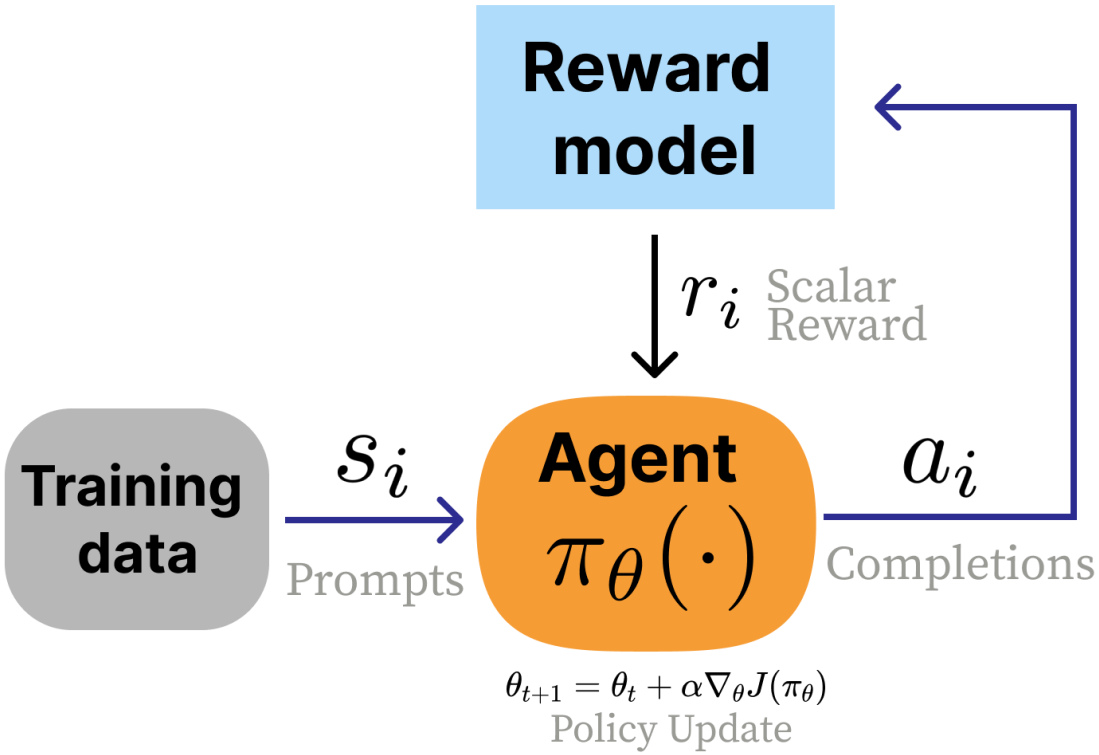
- Agent takes actions a_t in an environment with states s_t
- Reward is a known function $r(s_t, a_t)$ from the environment per step
- Optimize cumulative return over a trajectory (total steps T)

$$J(\pi) = \mathbb{E}_{\tau \sim \pi}\left[\sum_{t=0}^{T} \gamma^t r(s_t, a_t)\right]$$

RLHF

- No environment — prompts sampled from a dataset
- Reward is **learned** from human preferences (a proxy)
- **Response-level** reward (bandit-style, not per-token)
- Regularized with **KL penalty** to stay close to the base model

$$J(\pi) = \mathbb{E}\left[r_\theta(x, y)\right] - \beta\, D_{\mathrm{KL}}(\pi \| \pi_{\mathrm{ref}})$$



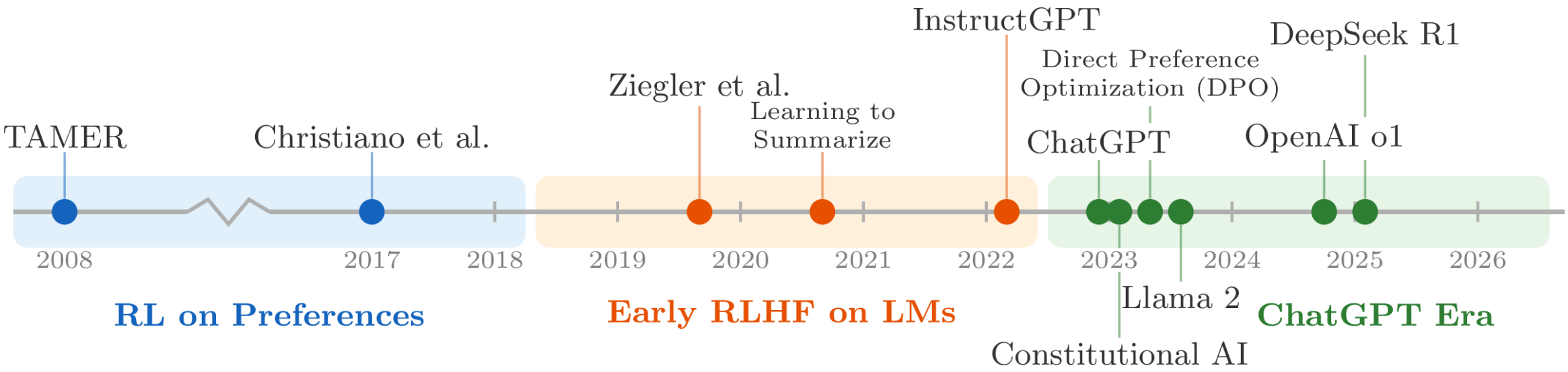
Christiano et al., 2017; Ouyang et al., 2022

Reinforcement learning with *Verifiable* rewards

Comparing classical RL vs. LLM RLHF and RLVR

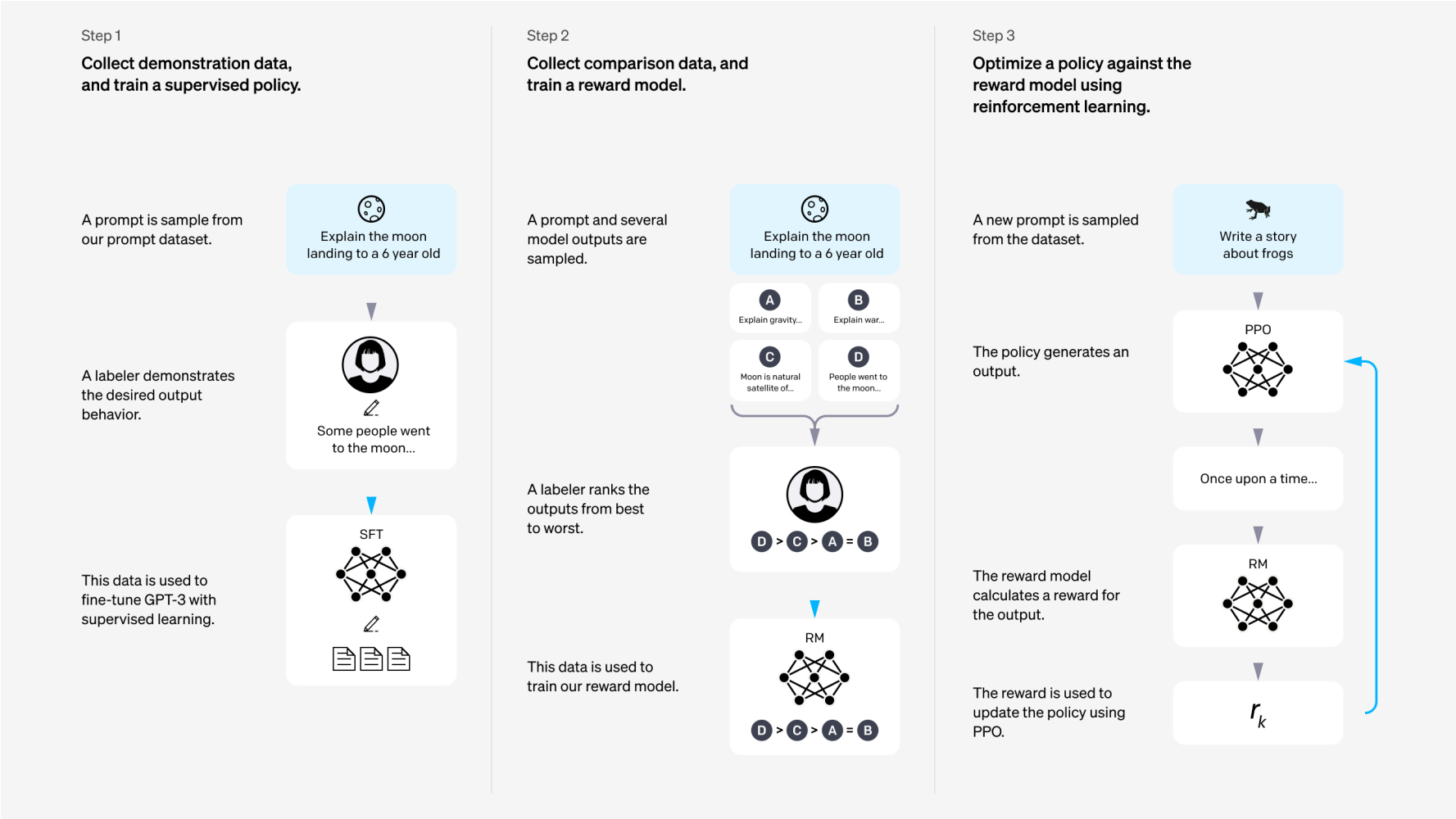
	Classical RL	RLHF	RLVR
Reward	Environment	Learned (proxy)	Verifiable (exact)
State transitions	Yes	No	No
Reward granularity	Per-step	Per-response	Per-response
Primary challenge	Explore-Exploit Trade-off	Over-optimization	Task generalization
Example	CartPole	Chat style tuning	Math reasoning

The path to modern RLHF



- **Ziegler 2019** (Ziegler et al., 2019) — first RLHF on language models
- **InstructGPT** (Ouyang et al., 2022) — the canonical RLHF recipe behind ChatGPT
- **Constitutional AI** (Bai et al., 2022) — Introduced early methods for AI feedback
- **DPO** (Rafailov et al., 2023) — direct preference optimization (DPO) without a reward model
- **Llama 3** (Grattafiori et al., 2024) and **Tülu 3** (Lambert et al., 2024) — modern multi-stage recipes

InstructGPT's 3-step RLHF recipe



The 3-step RLHF process figure from InstructGPT, which became "the standard" approach to RLHF for a few years.

Ouyang et al., 2022

Step 1/3: Instruction fine-tuning (IFT)

The foundation of post-training. Also called **Supervised Fine-tuning (SFT)**:

- Start from a pretrained language model
- Collect demonstrations of *desired* assistant behavior
- Train with standard supervised learning on prompt-response pairs. (different batch size, learning rate, etc.)
- Model can now answer questions. Easy to use IFT to quickly adapt base model to many domains.

SYSTEM

You are a helpful, harmless assistant. A system message like this can be used to steer the model to specific personas or behaviors.

USER

Write me a short poem about an optimistic goldfish.

ASSISTANT

Bright little goldfish
Finds a sunrise in each wave
Small bowl, endless hope

Ouyang et al., 2022

Step 2/3: Reward modeling

Overview:

- Collect **comparisons** between two model outputs for the same prompt
- RLHF gets its name from collecting *human* feedback between completions, but today much of it is AI feedback
- Train a reward model $r_\phi(x, y)$ to score preferred completions higher

The probability model says a response should win when it gets a higher reward score:

$$P(y_w \succ y_l \mid x) = \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$$

Training then minimizes the negative log-likelihood of the preferred response beating the rejected one:

$$\mathcal{L}_{\text{RM}}(\phi) = -\log \sigma(r_\phi(x, y_w) - r_\phi(x, y_l))$$

Notation:

- x is the prompt
- y_w is the **winning** response
- y_l is the **losing** response
- $r_\phi(x, y)$ is the trained reward model

Christiano et al., 2017; Ouyang et al., 2022

Step 2/3: Reward modeling

Core Idea

The reward used in RLHF is the model predicting the probability that a given piece of text would be the "winning" or "chosen" completion in a pair/batch. Clever!

The probability model says a response should win when it gets a higher reward score:

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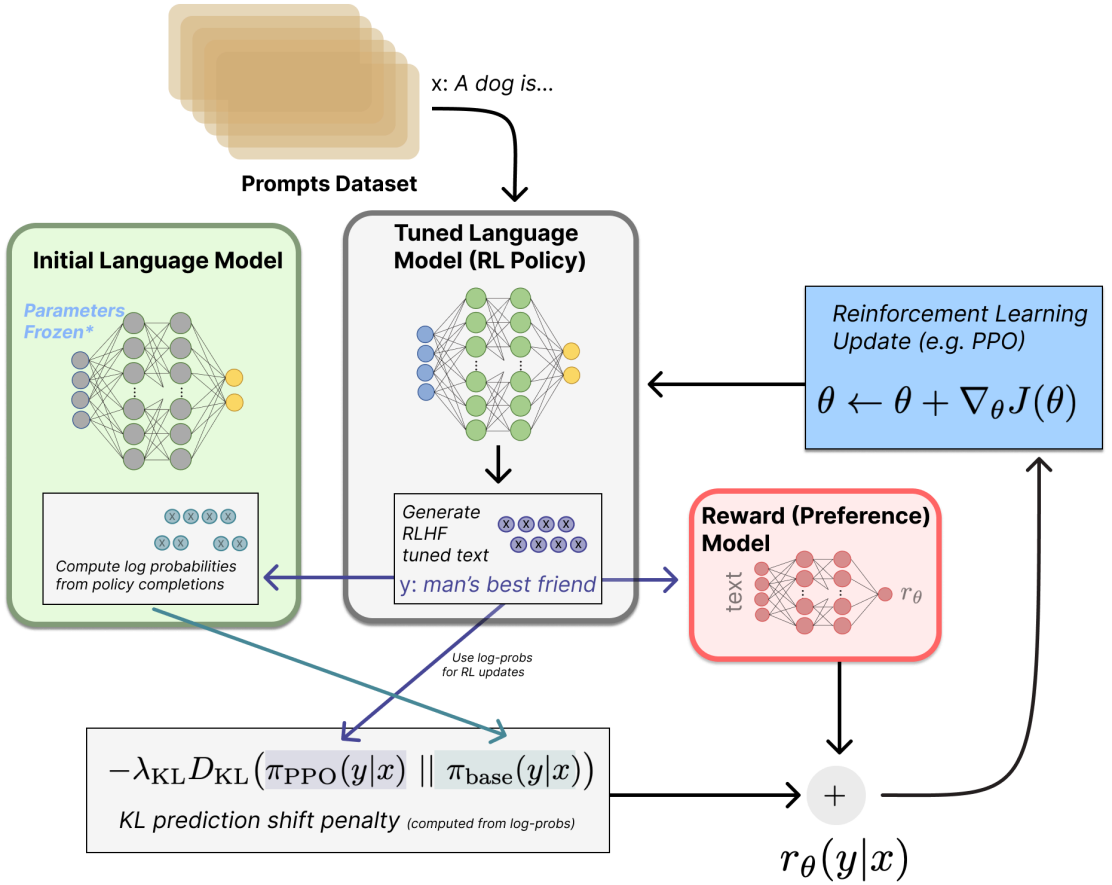
Christiano et al., 2017; Ouyang et al., 2022

Step 3/3: RL against the reward model

Where everything comes together (and RLHF gets its name):

- Sample a batch of prompts x_i from the dataset $\mathcal{D}$
- Generate completions $y_i \sim \pi_\theta(\cdot \mid x_i)$ from the model being trained
- Score them with the reward model $r_\phi(x_i, y_i)$
- Add a **KL penalty** so the policy stays close to the SFT/reference model.[1]
- Update the policy with a policy-gradient RL algorithm (Proximal Policy Optimization, PPO in InstructGPT & ChatGPT)

$$J(\pi) = \mathbb{E}[r_\phi(x, y)] - \beta D_{\mathrm{KL}}(\pi \,\|\, \pi_{\mathrm{ref}})$$



Ouyang et al., 2022

1: KL divergence measures how much the current policy differs from the reference model. For discrete outputs, $D_{\mathrm{KL}}(\pi \,\|\, \pi_{\mathrm{ref}}) = \mathbb{E}_{y\sim\pi}[\log \pi(y \mid x) - \log \pi_{\mathrm{ref}}(y \mid x)]$. People often colloquially call this the "KL distance" between the models, even though it is not a true metric.

The RLHF objective, unpacked

$$\max_{\pi} \; \mathbb{E}_{x\sim D,\, y\sim\pi(\cdot|x)} \quad \underbrace{r_\phi(x,y)}_{\text{maximize the reward}} \quad - \underbrace{\beta D_{\mathrm{KL}}\big(\pi(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big)}_{\text{but don't change the model too much}}$$

The reference model π_{ref} keeps the policy anchored to the SFT model.

$D_{\mathrm{KL}}\big(\pi(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x)\big)$ measures how far the new policy moves from that reference on prompt x.

β controls the tradeoff between **improving behavior** and **staying close** to what the model already knows.

What if we optimize this more directly?

$$\max_{\pi} \; \mathbb{E}_{x\sim D,\, y\sim\pi(\cdot|x)} r_\phi(x,y) - \beta D_{\mathrm{KL}}(\pi(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x))$$

Direct Preference Optimization (DPO)

- Derived the gradient toward the optimal solution, π^*, to the above equation
- Eliminated the need for a separate reward model (via training an implicit one)
- Train directly on preferred (y_w) vs. rejected (y_l) responses to a prompt (x)

$$\mathcal{L}_{\mathrm{DPO}}(\theta) = -\log \sigma \left(\beta \log \frac{\pi_\theta(y_w \mid x)}{\pi_{\mathrm{ref}}(y_w \mid x)} - \beta \log \frac{\pi_\theta(y_l \mid x)}{\pi_{\mathrm{ref}}(y_l \mid x)} \right)$$

Rafailov et al., 2023

What if we optimize this more directly?

$$\max_{\pi} \; \mathbb{E}_{x\sim D,\, y\sim\pi(\cdot|x)} r_\phi(x,y) - \beta D_{\mathrm{KL}}(\pi(\cdot \mid x) \,\|\, \pi_{\mathrm{ref}}(\cdot \mid x))$$

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DPO became very popular as it is

- Far simpler to implement
- Far cheaper to run
- Achieves ~80% or more of the final performance
- I used it to build models like Zephyr-Beta, Tülu 2/3, Olmo 2/3, etc.

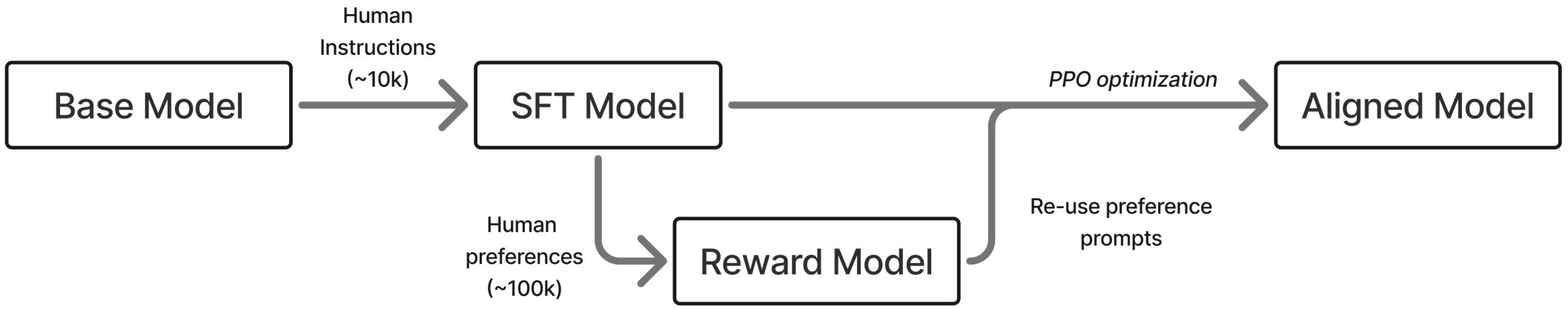
, 2023

How training recipes have evolved

	InstructGPT (2022)	Tülu 3 (2024)	DeepSeek R1 (2025)
Instruction data	~10K	~1M	100K+
Preference data	~100K	~1M	On-policy
RL stage	~100K prompts	~10K (RLVR)	N/A

An overall trend is to use far more compute across all the stages, but shifting more to RLVR.

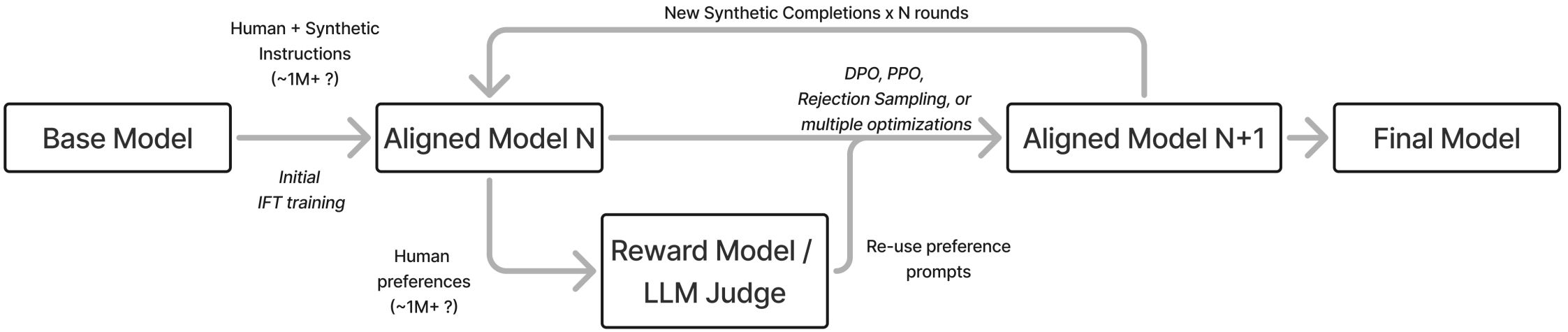
The early days: InstructGPT



Early on, RLHF had a well-documented, simple enough approach.

- **InstructGPT** made the classic three-stage recipe canonical: SFT, reward modeling, then RL against the reward model. *OpenAI even hinted that the original ChatGPT used this!*
- This became the intellectual template for much of modern post-training.

From RLHF to post-training



What began as an "RLHF" recipe evolved into a complex series of steps to get the final, best model (e.g. Nemotron 4 340B, Llama 3.1).

- Modern systems keep the same core idea of using multiple optimizers with different strengths and weaknesses, but add more stages, more data, and more filtering.
- This trend has only continued, and recipes ebb and flow, as tools like RLVR and model merging change the scope of what is doable in different ways.

From RLHF to “post-training”

As time has passed since ChatGPT, the field has gone through multiple distinct phases (roughly):

1. 2023: Simple SFT for better chatbots and reproducing RLHF fundamentals (Alpaca, Vicuna, etc.)
2. 2024: DPO dominates open models and training stages expand (Zephyr-beta, Tülu 2, etc.)
3. 2025: RLVR, complex recipes (Tülu 3, Olmo 3, Nemotron 3, R1, etc.)
4. 2026: Agentic training, multi-turn RL, etc.

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Within 2024 the field shifted its focus to post-training, as training stages evolved beyond the InstructGPT-style recipe, DPO proliferated, and largely RLHF was viewed as one tool (that you may not even need).

An intuition for post-training

RLHF's reputation was that its contributions are minor on the final language models.

> *"A model's knowledge and capabilities are learnt almost entirely during pretraining, while alignment teaches it which subdistribution of formats should be used when interacting with users."*

LIMA: Less Is More for Alignment (2023)

Zhou et al., 2023

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Sometimes this view of alignment (or RLHF) teaching "format" made people think that post-training only made minor changes to the model. This would describe finetuning as "*just style transfer*."

The base model trained on trillions of tokens of web text has seen and learned from an extremely broad set of examples. The model at this stage contains far more latent capability than early post-training recipes were able to expose.

The question is: How does post-training interact with these?

Muennighoff et al., 2025; Team, 2025; Zhou et al., 2023

An intuition for post-training

RLHF's reputation was that its contributions are minor on the final language models.

An example, **OLMoE** — same base model family, updated only post-training:

- `OLMoE-1B-7B-0924-Instruct` (Sep. 2024): **38.44** avg. eval score
- `OLMoE-1B-7B-0125-Instruct` (Jan. 2025): **45.62** avg. eval score

Base models determine the *ceiling*. Post-training's job has been to **reach it**.

Muennighoff et al., 2025; Team, 2025; Zhou et al., 2023

An intuition for post-training

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> *"A model's knowledge and capabilities are learnt almost entirely during pretraining, while alignment teaches it which subdistribution of formats should be used when interacting with users."*

LIMA: Less Is More for Alignment (2023)

> *"The superficial alignment hypothesis (SAH) posits that large language models learn most of their knowledge during pre-training, and that post-training merely surfaces this knowledge."*

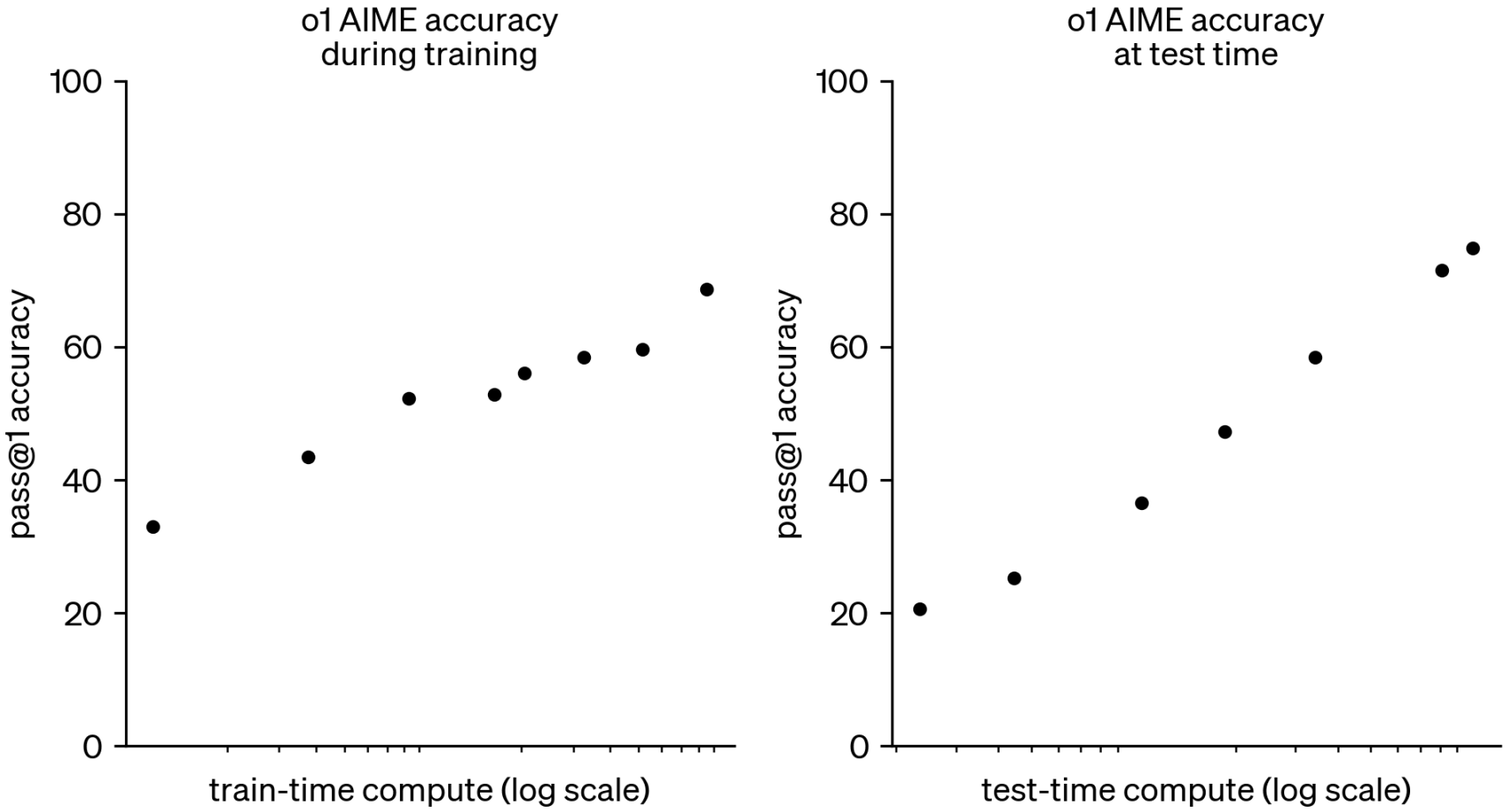
The second paper, 3 years later, matches my intuition for post-training.

I call this the Elicitation Theory of post-training, where we're trying to pull out the most useful knowledge of the model.

Vergara-Browne et al., 2026; Zhou et al., 2023

Beyond elicitation: The scaling RL era of post-training

OpenAI's seminal scaling plot with o1-preview

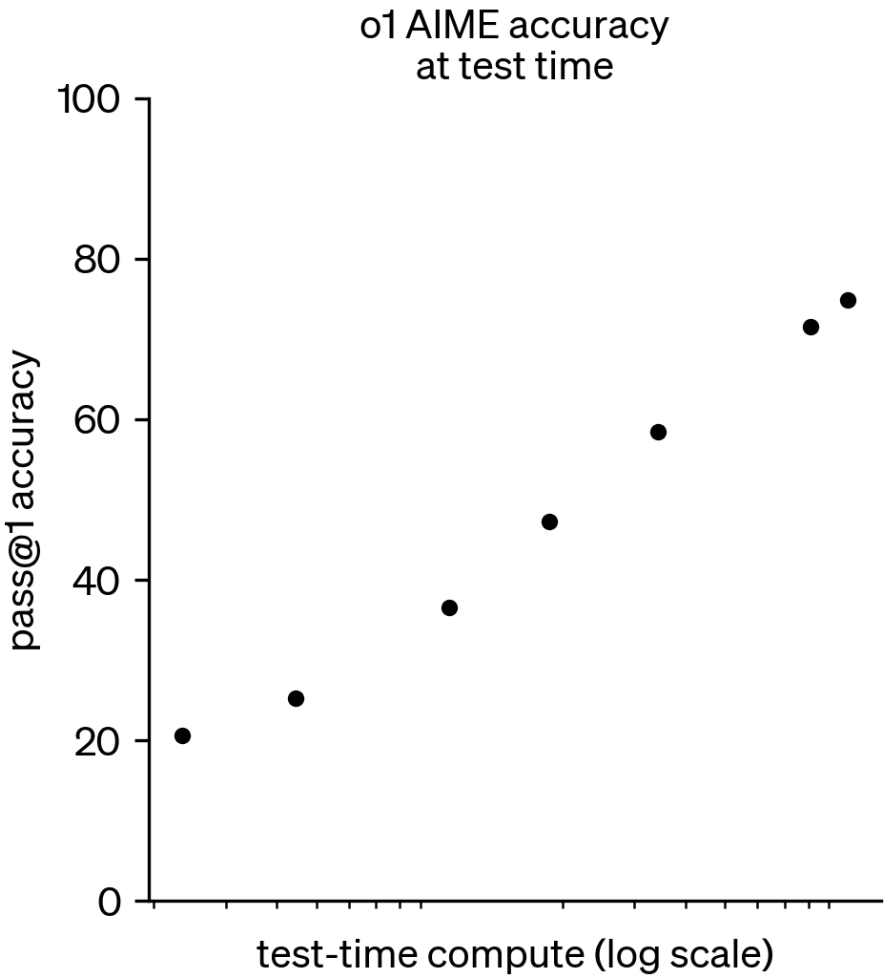


OpenAI, 2024

o1: Test-time scaling

A log-linear relationship between inference compute (number of tokens generated) and downstream performance.

- This is a fundamental property of models, unlocked in its popular form with RLVR
- Can be done in many ways: One long chain of thought (CoT) sequence, multiple agents in parallel, or mixes of the two
- Improving inference-time scaling changes the slope and offset of the curve



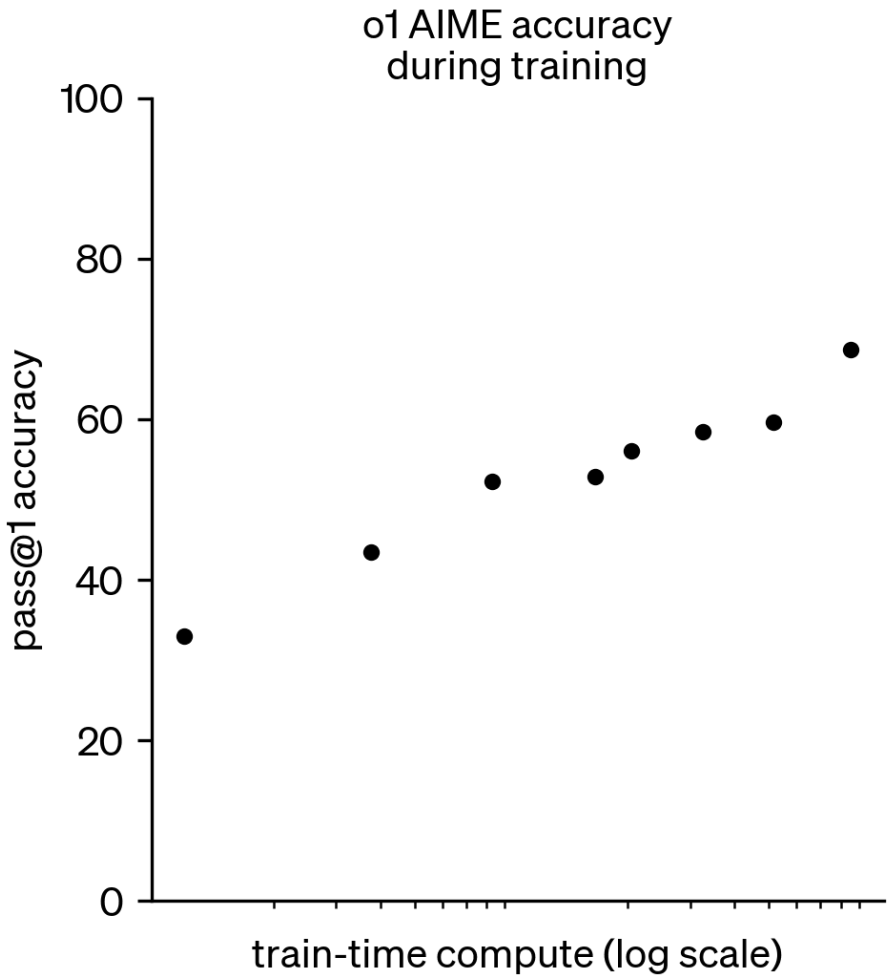
OpenAI, 2024

o1: Training-time scaling (with reinforcement learning!)

An often underplayed portion of the o1 release (and future reasoning/agentic models).

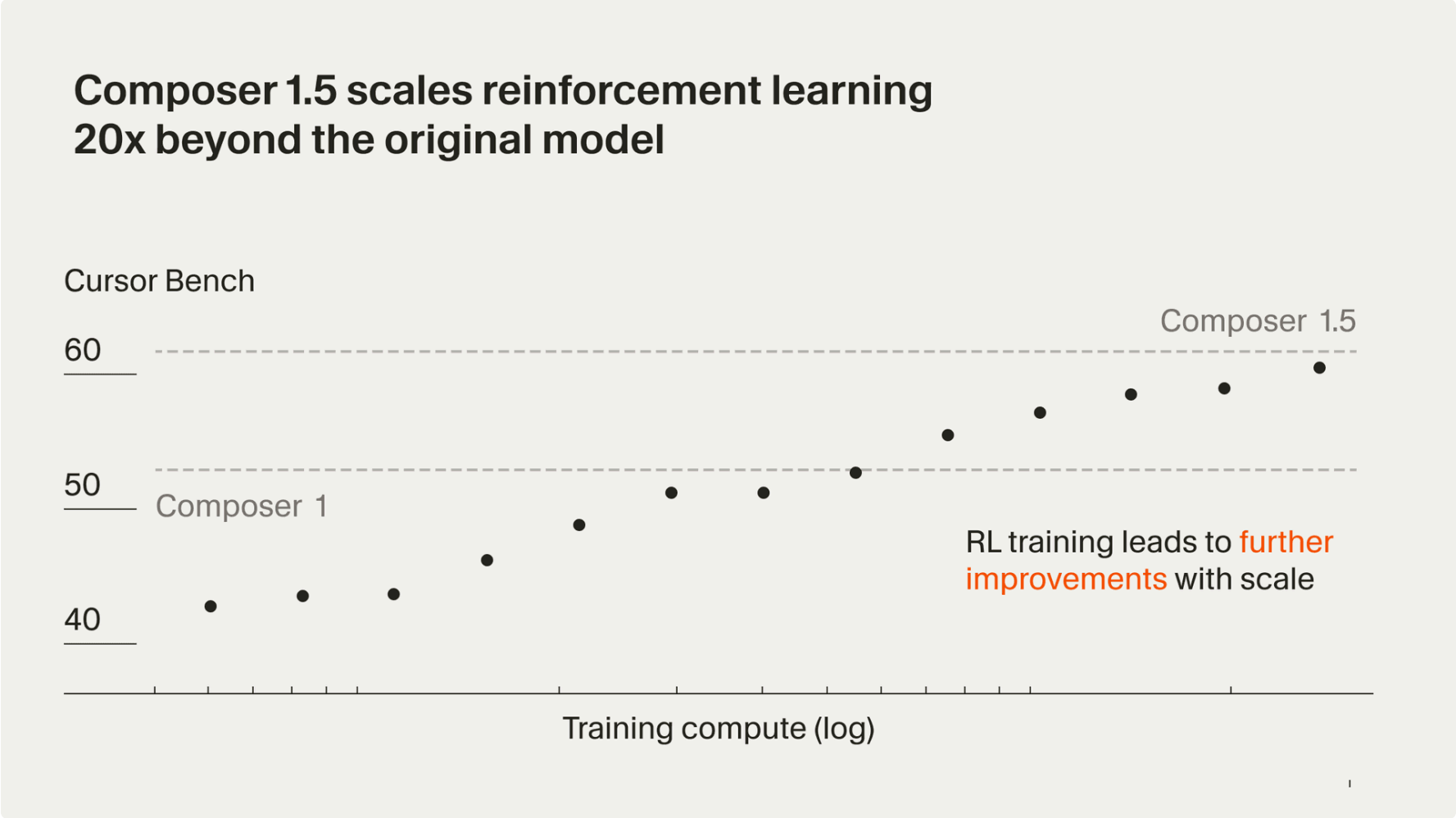
- Scaling reinforcement learning compute also has a log-linear return on performance!
- The core question: Is scaling RL *training* just eliciting more from the base model or actually teaching new abilities?

Results in a two-sided scaling landscape for training language models – both pretraining and post-training. The third place of scaling is at inference (no weight updates there).



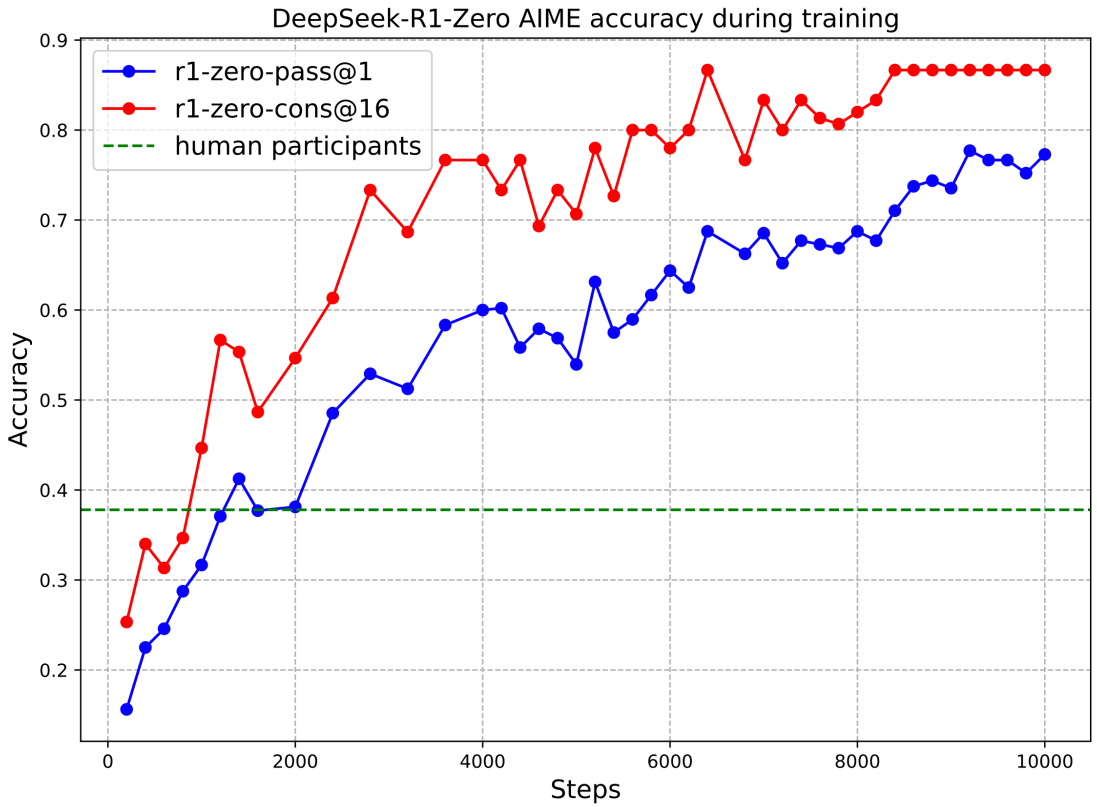
OpenAI, 2024

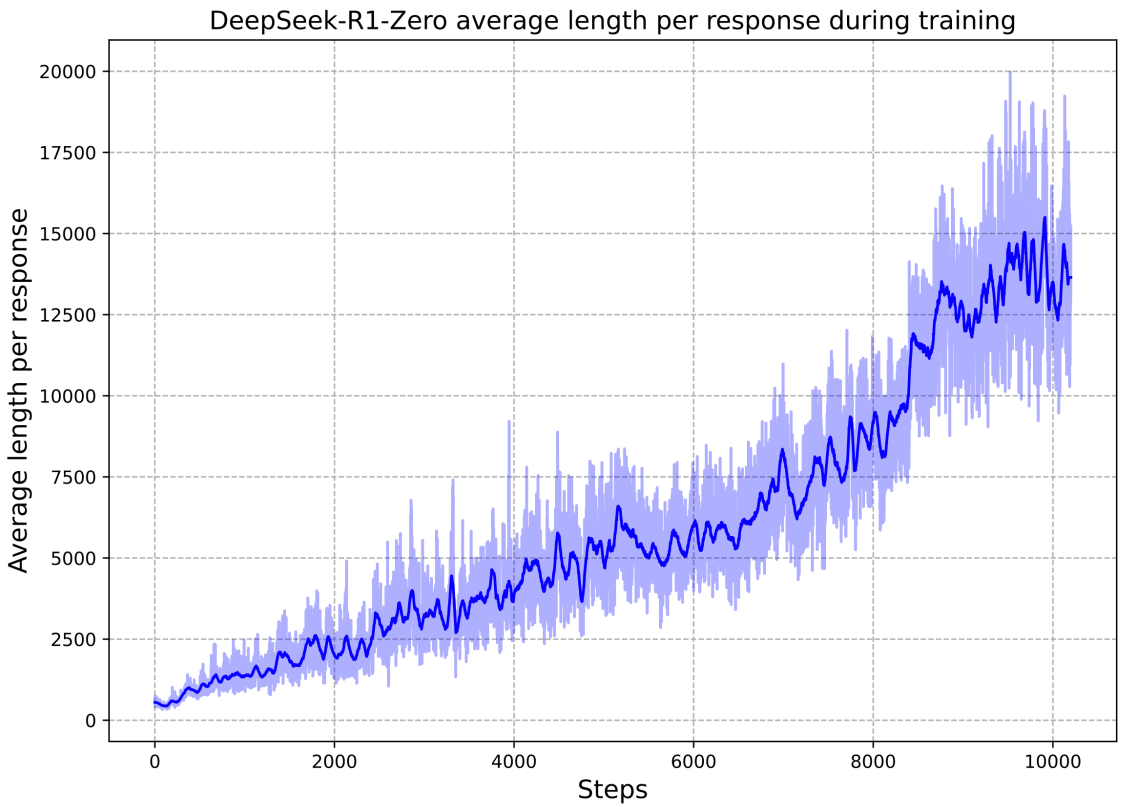
Cursor Composer 1.5: RL scaling



Team, 2026

DeepSeek-R1-Zero: RL scaling



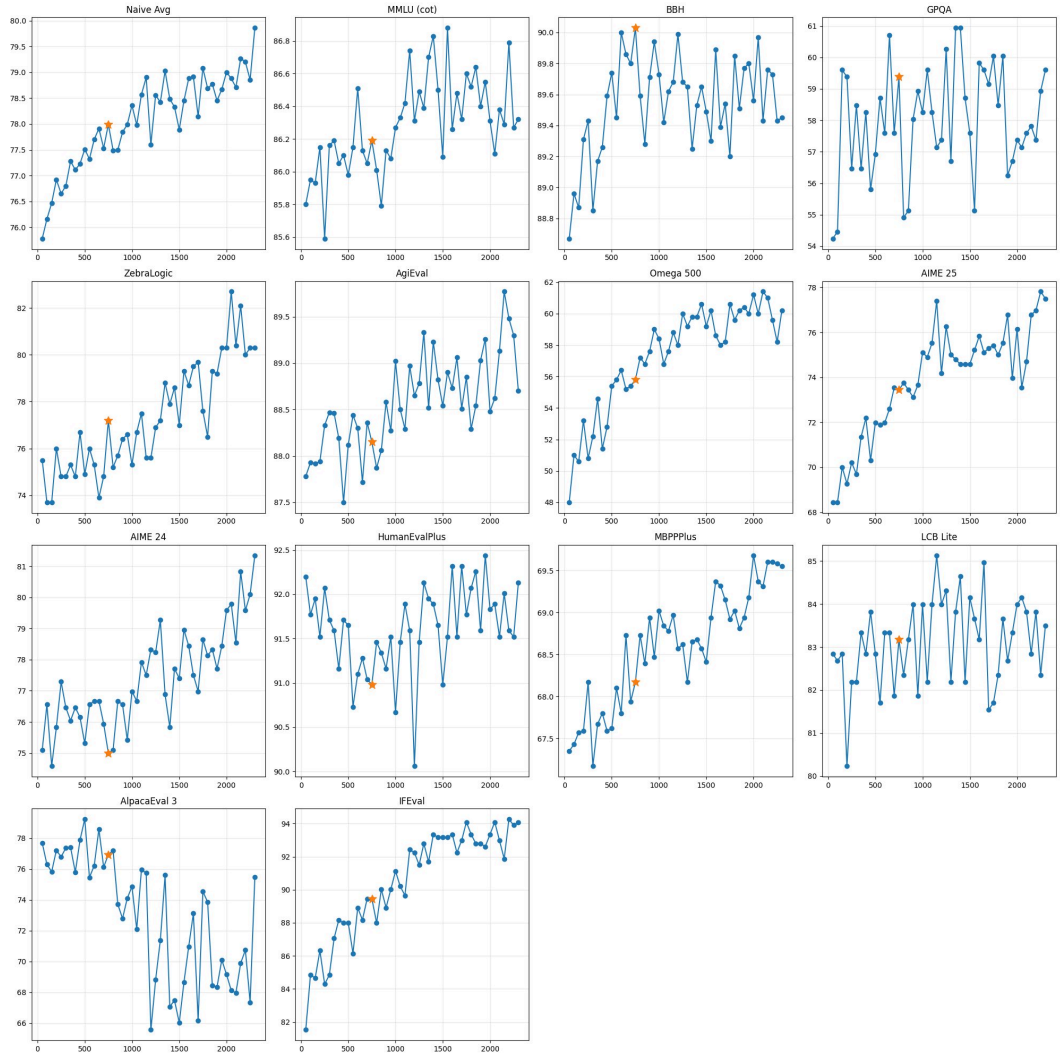


Guo et al., 2025

Olmo 3.1: extending the RL run

One of the few “fully open” large-scale RL runs to date.

- Training a general, 32B reasoning model.
- Full RL training took about **28 days on 224 GPUs**.
- Improvements in performance were very consistent across the run, in fact they were still going up when we had to stop it!



Olmo et al., 2025

Where this leaves us

Post-training and RLHF are changing faster than maybe ever before.

- Language models are becoming “tool-use native” and are now about tools, harnesses (how you tell the model to use said tools), and much more than just weights
- RLHF and human preferences haven’t gone away, but are evolving far more slowly and out of the central gaze of the industry
- Building language models and doing research is changing rapidly with coding agents

This talk is ~lecture 1 of a larger course

Introductions (this talk)

1. Introduction
2. Key Related Works
3. Training Overview

Core Training Pipeline

4. Instruction Tuning
5. Reward Models
6. Reinforcement Learning
7. Reasoning
8. Direct Alignment
9. Rejection Sampling

Data & Preferences

10. What are Preferences
11. Preference Data
12. Synthetic Data & CAI

Practical Considerations

13. Tool Use
14. Over-optimization
15. Regularization
16. Evaluation
17. Product & Character

Appendices

- A. Definitions
- B. Style & Information
- C. Practical Issues

Full lecture slides coming to rlhfbook.com/course and YouTube @natolambert!

Thank you

Sorry I could not make it in person!

Contact: nathan@natolambert.com

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rlhfbook.com



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